Experiment No. 1
Type of D.C machine

1. Object:

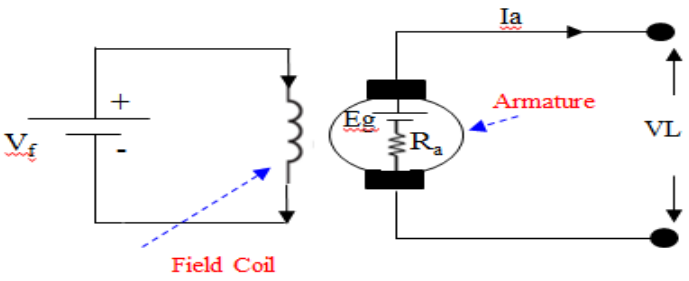
This experiment is done to teach the students the type of D.C machine (generators) and study of characteristic DC machine coils.

2. Theory:

Generators are usually classified the way in which their fields are excited. They are divided into:

A. Separately excited generator:

Generators are those whose field magnets are energized from an independent external source of D.C current. It is shown diagrammatically in the figure below.

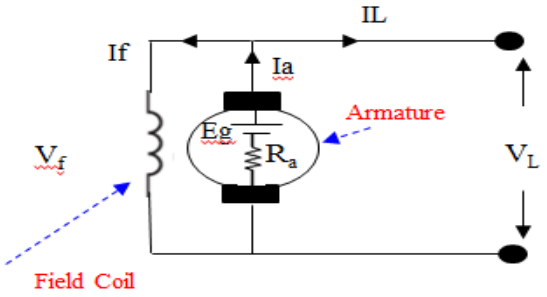


B. Self-excited:

Generator are those whose field magnets are energized by the current produced by the generators themselves .there are three types of self excited generator named according to the manner in which their field coil (or windings) are connected to the armature.

i. Shunt wound

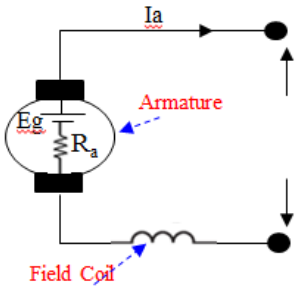
The field windings are connected across or in parallel with the armature conductors, and have the full voltage of the generator across them as shown in the figure.



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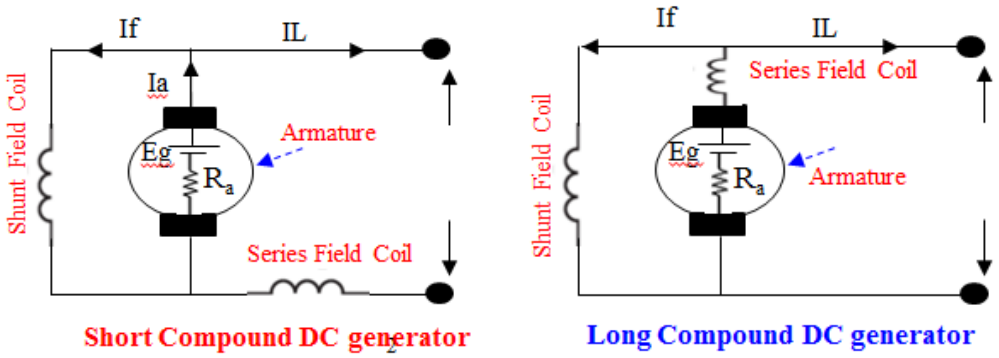
ii. Series wound

In this case, the field windings are jointed in series with the armature conductors as shown in the figure below.



ii. Compound wound

It is a combination of a few series and a few shunt windings and can be either short shunt or long shunt as shown in the figure below:



3. *Characteristics of D.C machine coils:*

A field coil is an electromagnet used to generate a magnetic field in an electro-magnetic machine, typically a rotating electrical machine such as a motor or generator. It consists of a coil of wire through which a current flows.

In a rotating machine, the field coils are wound on an iron magnetic core which guides the magnetic field lines. The magnetic core is in two parts; a stator which is stationary, and a rotor, which rotates within it. The magnetic field lines pass in a continuous loop or magnetic circuit from the stator through the rotor and back through the stator again. The field coils may be on the stator or on the rotor.

Hence the dc machine contains three type of coil:

1. Shunt field winding

In a DC shunt motor, the armature and field (shunt) windings are connected in parallel. A parallel circuit is also known as a shunt circuit; thus, the term, "shunt motor" is used. The shunt (field) windings of a DC shunt motor are ***made of smaller gauge wire,*** but they have ***many more turns than a series-wound*** DC motor.

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2. Series field winding

In a DC series motor, the armature and field (series) windings are connected in series. A series circuit is also known as a series motor; thus, the term, "series motor" is used. The series (field) windings of a DC series motor are ***made of thick gauge wire,*** but they have ***lower turns than a shunt-wound*** DC motor.

3. Armature winding

The **armature winding** is the main current-carrying **winding** in which the electromotive force or counter-emf of rotation is induced. The current in the **armature winding** is known as the **armature** current. ... In DC rotating machines other than brushless DC machines, it is usually rotating. The characteristics of the armature winding ***more turns than a series coil*** and ***lower turns than a shunt coil.***

$$R_f > R_a > R_s$$

There are three methods to determine the leads of the D.C machine coils:

1- By ohmmeter

Steps

1. Turn your multimeter dial to the lowest ohms setting, usually 20 or 200.



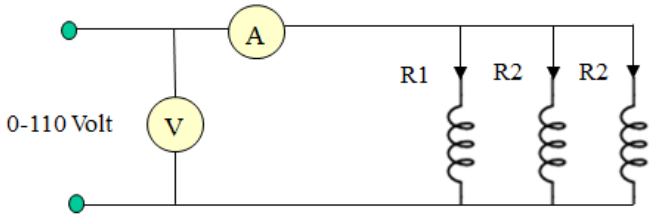
2. Touch the two probes together and take note of the internal resistance that the multimeter has. (Most multimeters have some degree of internal resistance).
3. Take note of this resistance reading high resistance is shunt field winding medium resistance armature winding and low resistance is series field resistance.

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2- By using D.C supply (ohms low)

Steps

1. Set the initial voltage to 0V and increase the voltage till Amper-miter reading half of rated current and measure the current and voltage, flowing through the circuit.
2. Now increase the voltage till the current arrived to rated current and again measure the current and voltage.



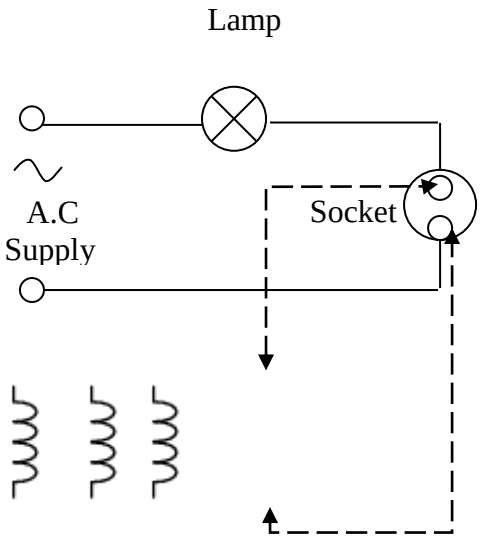
R1 = V1 / I1 R2 = V2 / I2 R3 = V3 / I3

3- By using the series circuit with lamp.

1- High fervency lamp (series field winding)

2- Low fervency lamp (shunt field winding)

3- Medium fervency lamp (armature winding)



Discussion:

1- What are the properties of each type (shunt, series, and armature coil)?
2- What the function of pole or pole shoes?
3- What the function of field coil?
4- Can you determine the leads of dc machine coils by using ac source? Why?
5- What is the purpose of yoke in DC machine?
6- What is the mean by self-exited and separately exited DC generator?

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